Linear Algebra — Lecture 4

Composition and Matrix Multiplication, Invertibility and Isomorphism

Friedberg–Insel–Spence, *Linear Algebra* 4e, §2.3, §2.4
D. C. Lay, *Linear Algebra and Its Applications* 4e, §2.1–§2.3

Topics. Composition of linear transformations, matrix multiplication, invertibility and isomorphism.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Composition	§2.1 (p. 94–95)	§2.3 (p. 86–87)
Matrix product	§2.1 (p. 95–96)	§2.3 (p. 87–88)
Matrix algebra	§2.1 (p. 93, 97–99)	§2.3 (p. 89–93)
Matrix of a composition	§2.1 (p. 94)	§2.3 (p. 88, 91)
Inverses, elementary matrices	§2.2 (p. 102–108)	§2.4 (p. 99–101)
The Invertible Matrix Theorem	§2.3 (p. 111–114)	§2.4 (p. 100–103)
Isomorphism and dimension	§2.3 (p. 114)	§2.4 (p. 102–105)

Two questions that drive the whole lecture: (1) *When two linear transformations are joined end to end, how do their matrices join?* (2) *What makes a transformation or a matrix reversible, and does reversibility change the dimension?*

How to read: use Lay first to get the algorithm under your fingers, then Friedberg to see why each step is legitimate.

All page numbers here are printed pages. For Lay, PDF page = printed page +17; for Friedberg, PDF page = printed page +15.

1 Composition of Linear Transformations

Definition 1.1 (Composition; Friedberg §2.3, p. 86)**.** Let V, W, Z be vector spaces over the same field $\mathbb{F}$, and let $T : V \to W$ and $U : W \to Z$ be linear. Joining the two **end to end** produces a new transformation from V to Z, written

$$UT : V \to Z, \qquad (UT)(x) = U\big(T(x)\big) \quad (x \in V),$$

the **composition** of T and U. The symbol UT is read "T first, then U": the T nearest to x acts first. Putting the earlier map on the right is a consequence of the function notation $(UT)(x) = U(T(x))$, not a separate convention.

Remark 1.1 (Reading a composition is where this lecture is most easily lost)**.** The hand writes U first, the computation does T first; the two orders are opposite. The cure is to read from the inside out: in $\big(U(T(x))\big)$ the inner T acts first. Composition is **associative**, $(UT)S = U(TS)$, whenever the maps fit together (this is associativity of function composition), so three or more maps may be bracketed freely; the **order**, however, cannot be changed: in general $UT \neq TU$.

Theorem 1.1 (A composition of linear maps is linear; Friedberg Theorem 2.9, §2.3, p. 86)**.** *If $T : V \to W$ and $U : W \to Z$ are linear, then $UT : V \to Z$ is linear.*

Proof. Let $x, y \in V$ and $c \in \mathbb{F}$. Then

$$\begin{aligned}(UT)(cx+y) &= U\big(T(cx+y)\big)\\ &= U\big(cT(x)+T(y)\big)\\ &= cU\big(T(x)\big)+U\big(T(y)\big) = c(UT)(x)+(UT)(y),\end{aligned}$$

where the second equality uses the linearity of T and the third the linearity of U. By the linearity test of Lecture 3 (T is linear exactly when $T(cx+y) = cT(x)+T(y)$ for all x, y, c), UT is linear. □

Remark 1.2 (The proof has nothing in it but "two uses of linearity")**.** The content of the previous proof is: send c and y through T, then send the result through U. Each step uses **the linearity of one map only**. This is why every algebraic law of composition can be checked pointwise: the properties of a composition come from linearity itself, not from any extra structure.

Theorem 1.2 (The algebra of $\mathcal{L}(V)$; Friedberg Theorem 2.10, §2.3, p. 87)**.** *Let V be a vector space, $T, U_1, U_2 \in \mathcal{L}(V)$, let $a \in \mathbb{F}$, and let I be the identity transformation on V. Then*

(a) $T(U_1+U_2) = TU_1+TU_2$, *and* $(U_1+U_2)T = U_1T+U_2T$;
(b) $T(U_1U_2) = (TU_1)U_2$;
(c) $TI = IT = T$;
(d) $a(U_1U_2) = (aU_1)U_2 = U_1(aU_2)$.

Remark 1.3 (Why it is stated for $\mathcal{L}(V)$ only; Theorem 1.2 and Friedberg Exercise 8, p. 96)**.** Every part of Theorem 1.2 needs two transformations that can be added or multiplied, and each of those operations has its own hypotheses: U_1+U_2 makes sense only when U_1, U_2 share both a domain and a codomain, and TU makes sense only when the domain of T is the codomain of U. Taking T, U_1, U_2 all in $\mathcal{L}(V)$ satisfies all of these at once, so all four identities can be written down. The general case ($T : V \to W$, $U : W \to Z$) is not false, it merely needs a different setup for each part, and Friedberg leaves it as Exercise 8. Checking first whether the symbols are even defined, and only then reading the equality, is the most useful habit in this section.

2 Matrix Multiplication

Definition 2.1 (The matrix product, by entries; Friedberg §2.3, p. 87)**.** Let A be an $m \times n$ matrix and B an $n \times p$ matrix (entries from $\mathbb{F}$). Their **product** AB is the $m \times p$ matrix whose (i, j)-entry is

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} \qquad (1 \le i \le m,\ 1 \le j \le p).$$

The only hypothesis is that the number of columns of A equals the number of rows of B; when that fails, AB is undefined and speaking of its entries is meaningless.

Definition 2.2 (The matrix product, by columns; Lay §2.1, p. 95)**.** Let A be an $m \times n$ matrix and $B = [\boldsymbol{b}_1 \ \cdots \ \boldsymbol{b}_p]$ an $n \times p$ matrix ($\boldsymbol{b}_j$ the j-th column of B). Then

$$AB = A[\boldsymbol{b}_1 \ \cdots \ \boldsymbol{b}_p] = [A\boldsymbol{b}_1 \ \cdots \ A\boldsymbol{b}_p],$$

that is, column j of AB is A applied to column j of B.

Definition 2.3 (The row–column rule; Lay §2.1, p. 96)**.** If AB is defined, then the entry in row i and column j of AB is the sum of the products of corresponding entries of row i of A and column j of B:

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Use this rule to compute by hand; use the sizes to decide whether the product exists: $(m \times n)(n \times p) = (m \times p)$, and the two inner n's must agree.

Remark 2.1 (The three descriptions are one thing)**.** Write column j of B as $\boldsymbol{b}_j$. By the column definition, column j of AB is $A\boldsymbol{b}_j$, and the i-th entry of $A\boldsymbol{b}_j$ is $\sum_k a_{ik}b_{kj}$, which is exactly the $(AB)_{ij}$ of the row–column rule. So the column definition and the entrywise definition give the same matrix; the row–column rule only looks at the same number from a different angle. Each description has its job: **entries** are for proofs (Theorem 2.7), **columns** are for the link with linear transformations (Theorem 2.1), and the **row–column rule** is for computation.

Theorem 2.1 (The columns of AB and the columns of B; Friedberg Theorem 2.13, §2.3, p. 90)**.** *Let A be $m \times n$ and B be $n \times p$, and let $\boldsymbol{u}_j$ and $\boldsymbol{v}_j$ denote the j-th columns of AB and of B. Then*

(a) $\boldsymbol{u}_j = A\boldsymbol{v}_j$;
(b) $\boldsymbol{v}_j = B\boldsymbol{e}_j$, where $\boldsymbol{e}_j$ is the j-th standard vector of $\mathbb{F}^p$.

Proof. (a) is the column definition written out.

(b) The i-th entry of $B\boldsymbol{e}_j$ is $\sum_k b_{ik}(\boldsymbol{e}_j)_k = b_{ij}$, and $(b_{1j}, \dots, b_{nj})^\top$ is precisely the j-th column $\boldsymbol{v}_j$ of B. □

Remark 2.2 (Reading (a) and (b) together)**.** Together the two parts of Theorem 2.1 say $(AB)\boldsymbol{e}_j = A(B\boldsymbol{e}_j)$: acting on a standard vector, matrix multiplication reproduces the composition "B first, then A". That is the motivation for the definition of the product and the germ of Theorem 2.3 below.

Remark 2.3 (Sparse matrices: left multiplication reads rows, right multiplication reads columns)**.** Let A be an $m \times n$ matrix with rows $\boldsymbol{r}_1, \dots, \boldsymbol{r}_m$ and columns $\boldsymbol{c}_1, \dots, \boldsymbol{c}_n$; let P be $m \times m$ and Q be $n \times n$. Then

$$\text{row } i \text{ of } PA = \sum_{k=1}^{m} P_{ik}\boldsymbol{r}_k, \qquad \text{column } j \text{ of } AQ = \sum_{k=1}^{n} Q_{kj}\boldsymbol{c}_k.$$

This is another way of writing the row-column rule. When P and Q have many zero entries they are called **sparse matrices**: no summation is then needed, and the product is read row by row and column by column. If the k-th entry of row i of P is 1 and the other entries of that row are 0, then row i of PA is row k of A; if that row has two 1s, in columns k and ℓ, then row i of PA is the sum of rows k and ℓ; if it has no 1, it is zero. Right multiplication is read column by column: if the k-th entry of column j of Q is 1, then column j of AQ is column k of A; two 1s give the sum of the two corresponding columns, and no 1 gives the zero column.

For example, $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ interchanges the two rows of A; for $P = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ row 1 of PA is row 2 of A and row 2 is zero, the 1 lying off the diagonal, so row 2 has been moved to row 1; $Q = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ makes column 1 of AQ the sum of the two columns of A, a column with two 1s adding two columns.

Theorem 2.2 (A matrix acting on coordinate vectors; Friedberg Theorem 2.14, §2.3, p. 91)**.** *Let V, W be finite-dimensional with ordered bases $\beta = \{v_1, \dots, v_n\}$ and $\gamma = \{w_1, \dots, w_m\}$, and let $T : V \to W$ be linear. Then for every $u \in V$,*

$$[T(u)]_\gamma = [T]_\beta^\gamma \, [u]_\beta.$$

Proof. Let $[u]_\beta = (a_1, \dots, a_n)^\top$, that is, $u = \sum_{i=1}^n a_i v_i$. By linearity, $T(u) = \sum_{i=1}^n a_i T(v_i)$. The coordinates of $T(v_i)$ in γ are column i of the matrix representation $[T]_\beta^\gamma$, so $T(v_i) = \sum_{j=1}^m [T]_{ji} w_j$ and

$$T(u) = \sum_{i=1}^{n} a_i \sum_{j=1}^{m} [T]_{ji} w_j = \sum_{j=1}^{m} \Big(\sum_{i=1}^{n} [T]_{ji} a_i \Big) w_j.$$

The bracket is entry j of $[T]_\beta^\gamma[u]_\beta$, and the coefficient of w_j on the left is entry j of $[T(u)]_\gamma$ by uniqueness of coordinates; hence the two agree. □

Theorem 2.3 (The matrix of a composition; Friedberg Theorem 2.11, §2.3, p. 88)**.** *Let V, W, Z be finite-dimensional with ordered bases α, β, γ, and let $T : V \to W$ and $U : W \to Z$ be linear. Then*

$$[UT]_\alpha^\gamma = [U]_\beta^\gamma\,[T]_\alpha^\beta.$$

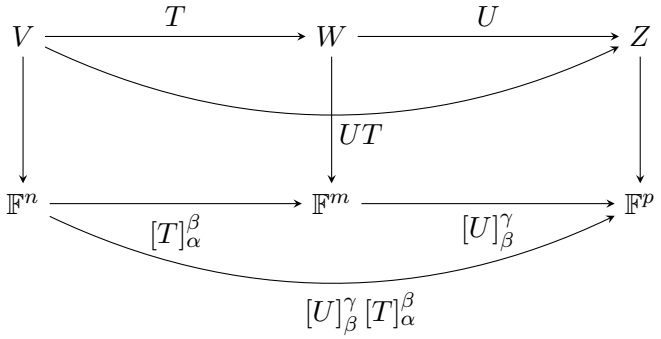
Proof. For every $x \in V$, apply Theorem 2.2 twice:

$$[UT(x)]_\gamma = [U\big(T(x)\big)]_\gamma = [U]_\beta^\gamma\,[T(x)]_\beta = [U]_\beta^\gamma\,[T]_\alpha^\beta\,[x]_\alpha.$$

On the other hand the same theorem applied to the composite UT gives $[UT(x)]_\gamma = [UT]_\alpha^\gamma\,[x]_\alpha$. Take $x = v_j$, the j-th vector of α (so $[x]_\alpha = \boldsymbol{e}_j$); the two displays then have the same j-th column. As j was arbitrary, the two matrices are equal. □

Proposition 2.4 (The case of one space; corollary to Theorem 2.3; Friedberg Theorem 2.11, §2.3, p. 89)**.** *Let V be finite-dimensional with ordered basis β, and let $T, U \in \mathcal{L}(V)$. Then $[UT]_\beta = [U]_\beta[T]_\beta$.*

Remark 2.4 (A commutative diagram: composition and matrix multiplication are one thing)**.**



The horizontal arrows are linear maps and their composites; the vertical arrows turn a vector into coordinates ($\phi_\alpha, \phi_\beta, \phi_\gamma$, which are isomorphisms by Theorem 4.9).

The top row is "T then U", the bottom row is "multiply by $[T]$, then by $[U]$". Theorem 2.3 says that this diagram **commutes**: walking along the top and then taking coordinates gives the same answer as taking coordinates first and then walking along the bottom. Read together with "matrix multiplication does not commute", it shows that the order in $[UT] = [U][T]$ is not a choice, it has to match the order of the composition.

Theorem 2.5 (Left-multiplication transformations; Friedberg Theorem 2.15, §2.3, p. 93)**.** *Let A be an $m \times n$ matrix with entries from $\mathbb{F}$, and let L_A be its left-multiplication transformation (Definition 2.2 of the Lecture 3 notes). Then L_A is linear, and for any $m \times n$ matrix B, with β, γ the standard ordered bases of $\mathbb{F}^n$ and $\mathbb{F}^m$:*

(a) $[L_A]_\beta^\gamma = A$;
(b) $L_A = L_B$ *if and only if* $A = B$;
(c) $L_{A+B} = L_A + L_B$ *and* $L_{aA} = aL_A$ *for all* $a \in \mathbb{F}$;
(d) if $T : \mathbb{F}^n \to \mathbb{F}^m$ *is linear, there is a* ***unique*** $m \times n$ *matrix* C *with* $T = L_C$; *in fact* $C = [T]_\beta^\gamma$;
(e) if E *is an* $n \times p$ *matrix, then* $L_{AE} = L_A L_E$;
(f) if $m = n$, *then* $L_{I_n} = I_{\mathbb{F}^n}$.

Remark 2.5 (What Theorem 2.5 says)**.** Parts (a) and (d) together say: **every linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$ is left multiplication by some matrix, and that matrix is unique**. This is the full version of the "standard matrix" of Lecture 3: in the standard bases, linear maps and matrices correspond one to one. Part (b) says $A \mapsto L_A$ is injective and (c) says

it preserves addition and scalar multiplication, so (a)–(c) make the correspondence bijective and operation-preserving. Part (e) is the interface between matrix multiplication and composition; the proof of associativity below uses only (e) and (b).

Theorem 2.6 (Associativity of matrix multiplication; Friedberg Theorem 2.16, §2.3, p. 93; Lay Theorem 2(a), §2.1, p. 97)**.** *Let $A(BC)$ be defined. Then $(AB)C$ is defined too, and $A(BC) = (AB)C$.*

Proof. By the definition of L_A, part (e) of Theorem 2.5 and associativity of function composition,

$$L_{A(BC)} = L_A L_{BC} = L_A(L_B L_C) = (L_A L_B)L_C = L_{AB} L_C = L_{(AB)C}.$$

Part (b) of Theorem 2.5 ($L_X = L_Y$ if and only if $X = Y$) then gives $A(BC) = (AB)C$. □

Theorem 2.7 (The algebra of matrices; Friedberg Theorem 2.12 and its corollary, §2.3, p. 89–90)**.** *Let A be $m \times n$, let B, C be $n \times p$, let D, E be $q \times m$, and let $a \in \mathbb{F}$. Then*

(a) $A(B + C) = AB + AC$, and $(D + E)A = DA + EA$;
(b) $a(AB) = (aA)B = A(aB)$;
(c) $I_m A = A = AI_n$ (note that the two identity matrices have different *sizes);*
(d) if V is an n-dimensional vector space with ordered basis β, then $[I_V]_\beta = I_n$.

Its corollary: for $B_1, \ldots, B_k$, $C_1, \ldots, C_k$ and scalars $a_1, \ldots, a_k$,

$$A\Big(\sum_{i=1}^{k} a_i B_i\Big) = \sum_{i=1}^{k} a_i A B_i, \qquad \Big(\sum_{i=1}^{k} a_i C_i\Big) A = \sum_{i=1}^{k} a_i C_i A$$

Proof. Parts (a) and (c) are entrywise computations. For instance

$$[A(B+C)]_{ij} = \sum_k a_{ik}(B+C)_{kj} = \sum_k (a_{ik}b_{kj} + a_{ik}c_{kj}) = \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj} = [AB]_{ij} + [AC]_{ij},$$

for all i, j, so the two matrices are equal; and $[I_m A]_{ij} = \sum_k (I_m)_{ik} a_{kj} = \sum_k \delta_{ik} a_{kj} = a_{ij}$, that is, $I_m A = A$, with the right-hand version proved the same way. The remaining parts are Exercise 5, p. 96. □

Theorem 2.8 (Transpose and products; Lay Theorem 3, §2.1, p. 99; Friedberg §2.3, p. 88)**.** *Let the sizes of A, B make the following operations defined, and let $r \in \mathbb{F}$. Then*

(a) $(A^\top)^\top = A$;
(b) $(A + B)^\top = A^\top + B^\top$;
(c) $(rA)^\top = rA^\top$;
(d) $(AB)^\top = B^\top A^\top$.

Proof. For (d), let A be $m \times n$ and B be $n \times p$. Entry (i, j) of $(AB)^\top$ is $(AB)_{ji} = \sum_k a_{jk} b_{ki}$, while entry (i, j) of $B^\top A^\top$ is $\sum_k (B^\top)_{ik} (A^\top)_{kj} = \sum_k b_{ki} a_{jk}$. The two sums differ only in the order of the factors, hence have the same value, so $(AB)^\top = B^\top A^\top$. The other parts are checked the same way (Friedberg proves only (d), in §2.3, p. 88; Lay's Theorem 3 gives all four). □

Remark 2.6 (Four wrong equations that look right; the warnings in Lay §2.1, p. 97–98)**.** This is where the algebra of matrices parts company with the algebra of numbers:

(1) $AB \neq BA$ in general (Lay's Example 7, p. 98 exhibits two products of the same pair);
(2) there is no cancellation law: $AB = AC$ does not force $B = C$;
(3) $AB = \mathbf{0}$ does not force $A = \mathbf{0}$ or $B = \mathbf{0}$;
(4) the transpose reverses the order: $(AB)^\top = B^\top A^\top$.

Note also what survives: distributivity, $A(B+C) = AB+AC$, holds exactly as for numbers. What fails is commutativity, cancellation, and "a product is zero only if a factor is".

3 Invertible Matrices

Definition 3.1 (Invertible matrix and inverse; Lay §2.2, p. 103; Friedberg §2.4, p. 100)**.** Let A be an $n \times n$ matrix. If there is an $n \times n$ matrix C with $AC = I_n$ and $CA = I_n$, then A is **invertible** and C is **an inverse** of A. An inverse, if it exists, is unique and is written A^{-1}. A square matrix with no inverse is **singular**; one that has an inverse is **nonsingular**.

Remark 3.1 (Why the definition writes both equations)**.** The definition demands $AC = I$ and $CA = I$ together, which looks stronger than "multiply out to an identity". For **square** matrices nothing is lost: if $AB = I$ with A, B of the same size, then both are invertible and each is the inverse of the other (Theorem 3.7). For non-square matrices that sentence fails at once, an $m \times n$ and an $n \times m$ matrix can multiply to an identity without either being called invertible. This is exactly why the Invertible Matrix Theorem lists "there is C with $CA = I$" and "there is D with $AD = I$" as two separate statements (parts (j) and (k) of Theorem 3.6).

Theorem 3.1 (The 2×2 inverse and the determinant test; Lay Theorem 4, §2.2, p. 103)**.** *Let* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$*. If* $ad - bc \neq 0$*, then* A *is invertible and*

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $ad - bc = 0$*, then* A *is not invertible.*

Remark 3.2 (What $ad - bc$ is)**.** The number $ad - bc$ is the **determinant** of A, written $\det A$; so Theorem 3.1 also says that a 2×2 matrix is invertible exactly when $\det A \neq 0$. The formula itself reads "swap a and d, negate b and c, divide by the determinant", and it holds for 2×2 only. It has an extension to order n: a later lecture introduces the determinant of order n, after which one sees that $ad - bc$ is exactly that object at $n = 2$. The $n \times n$ inverse is handled by Theorem 3.5 and the inversion algorithm.

Theorem 3.2 (Solving equations with an inverse; Lay Theorem 5, §2.2, p. 104)**.** *Let* A *be an invertible* $n \times n$ *matrix. Then for each* $\boldsymbol{b} \in \mathbb{R}^n$ *the equation* $A\boldsymbol{x} = \boldsymbol{b}$ *has the unique solution* $\boldsymbol{x} = A^{-1}\boldsymbol{b}$*.*

Proof. Existence: substituting $\boldsymbol{x} = A^{-1}\boldsymbol{b}$ gives $A(A^{-1}\boldsymbol{b}) = (AA^{-1})\boldsymbol{b} = \boldsymbol{b}$, by associativity and $AA^{-1} = I$. Uniqueness: if $A\boldsymbol{x}_1 = \boldsymbol{b} = A\boldsymbol{x}_2$, multiplying on the left by A^{-1} gives $\boldsymbol{x}_1 = \boldsymbol{x}_2$. □

Theorem 3.3 (The algebra of inverses; Lay Theorem 6, §2.2, p. 105)**.** *Let the matrices below be square of the same size and invertible. Then*

(a) $(A^{-1})^{-1} = A$*;*
(b) $(AB)^{-1} = B^{-1}A^{-1}$*, the inverse* ***reverses*** *the order;*
(c) $(A^{\top})^{-1} = (A^{-1})^{\top}$*.*

For more factors: a product of $n \times n$ *invertible matrices is invertible, and its inverse is the product of the inverses* ***in the reverse order*** *(boxed fact, Lay §2.2, p. 105).*

Proof. For (b), multiply $B^{-1}A^{-1}$ by AB on the left and on the right:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I, \qquad (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I,$$

using associativity twice; uniqueness of the inverse (Definition 3.1) gives $(AB)^{-1} = B^{-1}A^{-1}$. The proof shows why the order must be reversed: the inner factors can only cancel when B^{-1} sits next to B and A^{-1} next to A, and that is the only arrangement that does it. □

Definition 3.2 (Elementary matrix; Lay §2.2, p. 106)**.** An **elementary matrix** is a matrix obtained by performing a **single** elementary row operation on an identity matrix.

Proposition 3.4 (Elementary matrices and row operations; Lay §2.2, p. 106)**.**

(a) If an elementary row operation is performed on an $m \times n$ matrix A, the result is EA, where E comes from performing the same row operation on I_m;
(b) every elementary matrix is invertible, and E^{-1} is the elementary matrix of the same type that turns E back into I.

Theorem 3.5 (Invertible if and only if row equivalent to the identity; Lay Theorem 7, §2.2, p. 107)**.** *An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n, and in that case **any** sequence of elementary row operations that reduces A to I_n also turns I_n into A^{-1}.*

Proof. Suppose $A \sim I_n$. Each row operation is left multiplication by an elementary matrix (Proposition 3.4(a)), so there are elementary matrices $E_1, \dots, E_p$ with $E_p \cdots E_1 A = I_n$. Write $C = E_p \cdots E_1$; then $CA = I_n$, so by Theorem 3.7 A is invertible with $C = A^{-1}$. Hence $A^{-1} = E_p \cdots E_1 = E_p \cdots E_1 I_n$, which says that the same sequence of row operations applied to I_n produces A^{-1}.

Conversely, let A be invertible. By Theorem 3.2 the equation $A\boldsymbol{x} = \boldsymbol{b}$ has a solution for every $\boldsymbol{b}$, so every row of A has a pivot position; as A is square, the n pivots must lie on the diagonal and $A \sim I_n$. □

Algorithm 3.1 (The algorithm for finding A^{-1}; Lay §2.2, p. 108)**.**

Input: an $n \times n$ matrix A.
Output: A^{-1}, or the verdict that A is not invertible.

1. *write A beside I as $[\,A \mid I\,]$;*
2. *apply every row operation to both blocks at once, until the left block becomes I_n or a row of zeros appears;*
3. *if the left block becomes I_n, output the right block, which is A^{-1}; if a row of zeros appears, output the verdict that A is not invertible.*

Proof. One sequence of row operations is left multiplication by a single invertible matrix E, and $E[\,A \mid I\,] = [\,EA \mid E\,]$. When the left block becomes I_n we have $EA = I_n$, that is $E = A^{-1}$, so the right block is exactly A^{-1}. If the left block cannot become I_n, then A is not row equivalent to I_n, and Theorem 3.5 says A is not invertible. □

Theorem 3.6 (The Invertible Matrix Theorem; Lay Theorem 8, §2.3, p. 112)**.** *Let A be a **square** $n \times n$ matrix. Then the following statements are equivalent (for a given A they are either all true or all false):*

(a) A is an invertible matrix;
(b) A is row equivalent to I_n;
(c) A has n pivot positions;
(d) the equation $A\boldsymbol{x} = \boldsymbol{0}$ has only the trivial solution;
(e) the columns of A form a linearly independent set;
(f) the linear transformation $\boldsymbol{x} \mapsto A\boldsymbol{x}$ is one-to-one;
(g) the equation $A\boldsymbol{x} = \boldsymbol{b}$ has at least one solution for each $\boldsymbol{b} \in \mathbb{R}^n$;
(h) the columns of A span $\mathbb{R}^n$;
(i) the linear transformation $\boldsymbol{x} \mapsto A\boldsymbol{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$;
(j) there is an $n \times n$ matrix C with $CA = I$;
(k) there is an $n \times n$ matrix D with $AD = I$;
(l) $A^\top$ is invertible.

Remark 3.3 (Two hypotheses to watch when using the Invertible Matrix Theorem; Lay §2.3, p. 113)**.** The twelve statements describe **one** property, so only the cheapest one has to be checked: by hand, (b) or (c) (row reduce, count pivots); in an argument, (d) or (e) (trivial kernel, independent columns). Two things must be watched:

(1) the hypothesis is "A is an $n \times n$ *square* matrix". For a 4×3 matrix, "the columns are linearly independent" and "the columns span $\mathbb{R}^4$" can differ, and the whole table breaks (Lay §2.3, p. 113 warns about exactly this);
(2) parts (j) and (k) each require only **one** of the two equations. They may not be strengthened to $CA = AC = I$ on their own, what makes one side suffice is precisely the hypothesis that A is square (Remark 3.1).

Theorem 3.7 ($AB = I$ forces both factors to be invertible; boxed fact, Lay §2.3, p. 112)**.** *Let A, B be square matrices of the same size. If $AB = I$, then A and B are both invertible, with $B = A^{-1}$ and $A = B^{-1}$.*

Theorem 3.8 (Invertible linear transformations; Lay Theorem 9, §2.3, p. 114)**.** *Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be linear with standard matrix A. Then T is invertible if and only if A is an invertible matrix. In that case the linear transformation S given by $S(\boldsymbol{x}) = A^{-1}\boldsymbol{x}$ is the unique map satisfying $S(T(\boldsymbol{x})) = \boldsymbol{x}$ and $T(S(\boldsymbol{x})) = \boldsymbol{x}$.*

4 Invertible Linear Transformations and Isomorphisms

Definition 4.1 (Invertible linear transformation; Friedberg §2.4, p. 99; Lay §2.3, p. 113)**.** Let V, W be vector spaces and $T : V \to W$ linear. If there is $U : W \to V$ with $UT = I_V$ and $TU = I_W$ (the two identities on V and on W respectively), then T is **invertible** and U is **an inverse** of T. An inverse, if it exists, is unique and is written T^{-1}. In the language of sets: T is invertible exactly when T is a bijection.

Theorem 4.1 (The inverse of a linear map is linear; Friedberg Theorem 2.17, §2.4, p. 100)**.** *If $T : V \to W$ is linear and invertible, then $T^{-1} : W \to V$ is linear.*

Proof. Let $y_1, y_2 \in W$ and $c \in \mathbb{F}$, and write $x_1 = T^{-1}(y_1)$, $x_2 = T^{-1}(y_2)$, so that $T(x_1) = y_1$ and $T(x_2) = y_2$. By linearity of T, $T(cx_1 + x_2) = cy_1 + y_2$; applying T^{-1} to both sides gives $T^{-1}(cy_1 + y_2) = cx_1 + x_2 = cT^{-1}(y_1) + T^{-1}(y_2)$, so T^{-1} is linear. □

Proposition 4.2 (An invertible map preserves dimension; Lemma in Friedberg §2.4, p. 101)**.** *Let $T : V \to W$ be an invertible linear transformation. Then V is finite-dimensional if and only if W is finite-dimensional; in that case* $\dim V = \dim W$.

Proof. Assume V is finite-dimensional with basis $\beta = \{x_1, \dots, x_n\}$. The range is the span of the images of a basis (Lecture 3, $\mathcal{R}(T) = \operatorname{Span} T(\beta)$), and T is onto because it is invertible, so $W = \operatorname{Span}\{T(x_1), \dots, T(x_n)\}$; hence W is finite-dimensional. Conversely, if W is finite-dimensional, the same argument applied to T^{-1} (linear by Theorem 4.1) makes V finite-dimensional.

Suppose now that both are finite-dimensional. The vectors $T(x_1), \dots, T(x_n)$ span W; and since T is one-to-one, the independence of $x_1, \dots, x_n$ gives the independence of $T(x_1), \dots, T(x_n)$ (Lecture 3: T is one-to-one exactly when its kernel is trivial). Hence they form a basis of W and $\dim W = n = \dim V$. □

Theorem 4.3 (Invertible if and only if the matrix representation is invertible; Friedberg Theorem 2.18, §2.4, p. 101)**.** *Let V, W be finite-dimensional with ordered bases β and γ, and let $T : V \to W$ be linear. Then T is invertible if and only if $[T]_\beta^\gamma$ is invertible; in that case*

$$[T^{-1}]_\gamma^\beta = \left([T]_\beta^\gamma\right)^{-1}.$$

Proof. Assume T is invertible. By Proposition 4.2, $\dim V = \dim W = n$, so $A = [T]_\beta^\gamma$ is $n \times n$. From $T^{-1}T = I_V$, $TT^{-1} = I_W$ and Theorem 2.3,

$$I_n = [I_V]_\beta = [T^{-1}T]_\beta = [T^{-1}]_\gamma^\beta [T]_\beta^\gamma, \qquad I_n = [TT^{-1}]_\gamma = [T]_\beta^\gamma [T^{-1}]_\gamma^\beta,$$

so $[T^{-1}]_\gamma^\beta$ is an inverse of A; hence A is invertible and $[T^{-1}]_\gamma^\beta = A^{-1}$.

Conversely, let $A = [T]_\beta^\gamma$ be invertible, with $\dim V = \dim W = n$. Write $\beta = \{v_1, \ldots, v_n\}$. An invertible matrix has linearly independent columns (part (e) of the Invertible Matrix Theorem). If $x = \sum_i a_i v_i$ satisfies $T(x) = \mathbf{0}$, then Theorem 2.2 gives $A[x]_\beta = \mathbf{0}$, hence $[x]_\beta = \mathbf{0}$ and $x = \mathbf{0}$: the kernel is trivial and T is one-to-one. Between spaces of equal finite dimension a one-to-one map is onto (Lecture 3: $\dim V = \dim W$ and $\operatorname{rank}(T) = \dim V$ force surjectivity), so T is a bijection, hence invertible. □

Proposition 4.4 (Two special cases; corollaries to Theorem 4.3; Friedberg Theorem 2.18, §2.4, p. 102)**.**

(1) Let V be finite-dimensional with ordered basis β, and let $T \in \mathcal{L}(V)$. Then T is invertible if and only if $[T]_\beta$ is, and $[T^{-1}]_\beta = ([T]_\beta)^{-1}$.

(2) Let A be an $n \times n$ matrix. Then A is invertible if and only if L_A is invertible, and $(L_A)^{-1} = L_{A^{-1}}$.

Definition 4.2 (Isomorphism; Friedberg §2.4, p. 102)**.** Let V, W be vector spaces over $\mathbb{F}$. If there is an invertible linear transformation $T : V \to W$, then V is **isomorphic to** W (written $V \cong W$), and T is called an **isomorphism** from V to W.

Remark 4.1 (Invertible and isomorphic differ in what they are about)**.** "Invertible" is a property of **one map**: it can be undone (equivalently, it is a bijection). "Isomorphic" is a relation between **two spaces**: there exists an invertible linear map carrying one onto the other. Strictly speaking, $T : V \to W$ is invertible exactly when T is an isomorphism from V to W, and "V is isomorphic to W" is the conclusion of that sentence. An invertible linear map of a space to itself is sometimes called an **automorphism**. When a problem asks whether T is an isomorphism, first fix which two spaces are involved; when it asks whether V and W are isomorphic, either exhibit a map or quote the dimension criterion.

Theorem 4.5 (The isomorphism criterion; Friedberg Theorem 2.19, §2.4, p. 103)**.** *Let V, W be finite-dimensional vector spaces over the* ***same field****. Then V is isomorphic to W if and only if $\dim V = \dim W$.*

Proof. Necessity is Proposition 4.2: an isomorphism is an invertible linear map, and such a map preserves dimension.

For sufficiency, let $\dim V = \dim W = n$ and choose bases $\beta = \{v_1, \ldots, v_n\}$ of V and $\gamma = \{w_1, \ldots, w_n\}$ of W. By "a linear map is determined by its values on a basis" (Lecture 3), there is a linear $T : V \to W$ with $T(v_i) = w_i$ for $i = 1, \ldots, n$. Since the range is the span of the images of a basis, $\mathcal{R}(T) = \operatorname{Span} T(\beta) = \operatorname{Span} \gamma = W$, so T is onto; and the dimension theorem $\dim \mathcal{R}(T) = \dim V - \dim \mathcal{N}(T)$ together with $\dim \mathcal{R}(T) = n = \dim V$ gives $\dim \mathcal{N}(T) = 0$, so T is one-to-one. Thus T is a bijection and $V \cong W$. □

Proposition 4.6 (Isomorphic to a coordinate space; corollary to Theorem 4.5; Friedberg Theorem 2.19, §2.4, p. 103)**.** *Let V be a vector space over $\mathbb{F}$. Then V is isomorphic to $\mathbb{F}^n$ if and only if $\dim V = n$.*

Theorem 4.7 (The space of linear maps is isomorphic to a matrix space; Friedberg Theorem 2.20, §2.4, p. 103)**.** *Let V, W be finite-dimensional over $\mathbb{F}$ with $\dim V = n$, $\dim W = m$, and let β, γ be ordered bases. Then the map $\Phi : \mathcal{L}(V, W) \to M_{m \times n}(\mathbb{F})$ defined by $\Phi(T) = [T]_\beta^\gamma$ is an isomorphism.*

Proof. By Definition 3.4 and Proposition 3.2(b) of the Lecture 3 notes, $\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$ under the pointwise operations.

Φ is linear: by Theorem 2.2, for every x, $[(T+U)(x)]_\gamma = [T(x)]_\gamma + [U(x)]_\gamma = [T]^\gamma_\beta [x]_\beta + [U]^\gamma_\beta [x]_\beta$; running x through β and comparing columns gives $[T+U]^\gamma_\beta = [T]^\gamma_\beta + [U]^\gamma_\beta$, and scalar multiplication is the same (it also follows from the corollary to Theorem 2.7).

Φ is injective: if $[T]^\gamma_\beta = 0$, then every $T(v_i)$ has zero γ-coordinates, so $T(v_i) = \mathbf{0}$, and "determined by its values on a basis" gives $T = 0$.

Φ is surjective: given $A = (a_{ij}) \in M_{m\times n}(\mathbb{F})$, prescribe T on the basis by $T(v_j) = \sum_i a_{ij} w_i$; the same theorem produces a linear T with $[T]^\gamma_\beta = A$ by definition of the matrix representation. Hence Φ is a bijection, that is, an isomorphism. □

Proposition 4.8 (The dimension of a space of linear maps; corollary to Theorem 4.7; Friedberg Theorem 2.20, §2.4, p. 104)**.** *Let V, W be finite-dimensional with $\dim V = n$ and $\dim W = m$. Then $\mathcal{L}(V, W)$ is finite-dimensional and $\dim \mathcal{L}(V, W) = mn$.*

Proof. By Theorem 4.7 and the isomorphism criterion (Theorem 4.5), $\mathcal{L}(V, W)$ and $M_{m\times n}(\mathbb{F})$ have the same dimension; a basis of $M_{m\times n}(\mathbb{F})$ is the family of matrix units E_{ij} (Lecture 2), of which there are mn. Hence $\dim \mathcal{L}(V, W) = mn$. □

Definition 4.3 (Standard representation; Friedberg §2.4, p. 104)**.** Let β be an ordered basis for an n-dimensional vector space V over $\mathbb{F}$. The map that turns a vector into coordinates,

$$\phi_\beta : V \to \mathbb{F}^n, \qquad \phi_\beta(x) = [x]_\beta,$$

is the **standard representation** of V with respect to β.

Theorem 4.9 (The standard representation is an isomorphism; Friedberg Theorem 2.21, §2.4, p. 104)**.** *For any finite-dimensional vector space V with an ordered basis β, ϕ_β is an isomorphism.*

Proof. Linearity is the statement $[cx + y]_\beta = c[x]_\beta + [y]_\beta$, which is the definition of coordinates plus uniqueness of the representation in a basis. Injectivity: if $[x]_\beta = \mathbf{0}$, then $x = \sum_i 0 \cdot v_i = \mathbf{0}$ (unique representation). Surjectivity: every $(a_1, \ldots, a_n)^\top \in \mathbb{F}^n$ is the coordinate vector of $\sum_i a_i v_i$. So ϕ_β is a linear bijection, that is, an isomorphism. □

Remark 4.2 (Two correspondences that must not be confused)**.** The lecture contains two one-to-one correspondences; keeping them apart prevents most mistakes:

(1) Theorem 2.5: linear transformations $\mathbb{F}^n \to \mathbb{F}^m$ correspond to $m \times n$ matrices, the correspondence being "take the matrix in the standard bases". This is why every linear system can be written $A\boldsymbol{x} = \boldsymbol{b}$;
(2) Theorem 4.7: in general $\mathcal{L}(V, W)$ is isomorphic to $M_{m\times n}(\mathbb{F})$, and the correspondence depends on the **chosen** ordered bases β, γ. Change the bases and the matrix of the same T changes (the subject of Lecture 5).

5 A Worked Example

Example 5.1 (Composition, inversion and isomorphism; this example was written for this lecture; the methods are Friedberg §2.3–§2.4 and Lay §2.1–§2.3)**.** Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have standard matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix},$$

and let $U : \mathbb{R}^3 \to \mathbb{R}^3$ have as standard matrix the matrix B obtained by interchanging rows 1 and 3 of I_3. Find the standard matrices of UT and of TU and explain why they differ in general; find A^{-1} by row reducing $[\,A \mid I_3\,]$; solve $A\boldsymbol{x} = \boldsymbol{b}$ for $\boldsymbol{b} = (1, 0, 1)^\top$; finally explain why T is an isomorphism, write down $[T^{-1}]$, and use $(UT)^{-1} = T^{-1}U^{-1}$ to check the earlier results.

Solution. **(1) Composing in both orders.** The matrix B is the elementary matrix that interchanges rows 1 and 3 of I_3:

$$B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

By Theorem 2.3,

$$[UT] = [U][T] = BA = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 6 & 0 \\ 0 & 1 & 4 \\ 1 & 2 & 3 \end{bmatrix},$$

$$[TU] = [T][U] = AB = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 1 & 0 \\ 0 & 6 & 5 \end{bmatrix}.$$

The difference can be read off word by word: left multiplication by B interchanges the first and third *rows* of A (this is Proposition 3.4(a): left multiplication by an elementary matrix performs one row operation), whereas right multiplication by B interchanges the first and third *columns*. Composition is not commutative (see Remark 2.6), so $BA \neq AB$.

(2) Finding A^{-1}. Place I_3 beside A and row reduce the whole array:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 5 & 6 & 0 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 16 & -12 & -3 \\ 0 & 1 & 0 & 20 & -15 & -4 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -24 & 18 & 5 \\ 0 & 1 & 0 & 20 & -15 & -4 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array}\right].$$

(One step, dividing row 3 by -15, is omitted here.) The left block has become I_3, so A is invertible (Theorem 3.5, that is, part (b) of the Invertible Matrix Theorem), and

$$A^{-1} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}.$$

(3) Solving $A\boldsymbol{x} = \boldsymbol{b}$. By Theorem 3.2,

$$\boldsymbol{x} = A^{-1}\boldsymbol{b} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -19 \\ 16 \\ -4 \end{bmatrix}.$$

Check by substituting: the three rows of A give $-19 + 32 - 12 = 1$, $16 - 16 = 0$, $-95 + 96 = 1$, which match $\boldsymbol{b} = (1, 0, 1)^\top$.

(4) Isomorphism and the inverse transformation. A is invertible, so T is invertible (Theorem 3.8; or read it off in the standard bases from Theorem 4.3). Hence T is an automorphism of $\mathbb{R}^3$, and

$$[T^{-1}] = [T]^{-1} = A^{-1} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}. \qquad \square$$

Approach. Three tools cover the three parts. The order of a composition is fixed by Theorem 2.3: $[UT] = [U][T]$, so "T then U" puts $[U]$ on the left. Inversion needs no 2×2 formula, place I beside A and row reduce (Algorithm 3.1), the test being whether A becomes I_n. Isomorphism is a matter of counting dimensions (Theorem 4.5): $\dim \mathbb{R}^3 = \dim \mathbb{R}^3$ and T is invertible, so T is an automorphism of $\mathbb{R}^3$ and $[T^{-1}]$ is A^{-1}. The last step deliberately uses the algebra of inverses, to see the reversal once more.

6 Checklist and Exercises

By the end of this lecture you should be able to:

1. state why the order of writing UT and the order of computing it are opposite, and write out the two layers of $(UT)(x)$;
2. prove that a composition of linear maps is linear using "two uses of linearity", saying which step uses T and which uses U;
3. write down both the entrywise and the column definition of the matrix product, and explain why they agree (the row–column rule);
4. state every hypothesis of Theorem 2.3 (finite-dimensional, ordered bases) and write $[UT]_\alpha^\gamma$ together with the corollary $[UT]_\beta = [U]_\beta[T]_\beta$;
5. use Theorem 2.5 to explain "linear maps $\mathbb{F}^n \to \mathbb{F}^m$ correspond to $m \times n$ matrices", and say how the correspondence is made;
6. explain why associativity can be proved with left-multiplication transformations, and which step uses $L_A = L_B \Rightarrow A = B$;
7. compare the direction of $(AB)^{-1} = B^{-1}A^{-1}$ with that of $(AB)^\top = B^\top A^\top$, and explain why both reverse;
8. write the 2×2 inverse formula, say what $ad - bc$ is, and state the inversion algorithm $[\,A \mid I\,] \sim [\,I \mid A^{-1}\,]$;
9. list at least six equivalent statements of the Invertible Matrix Theorem and point out that the theorem needs a square matrix;
10. state both hypotheses of the isomorphism criterion (Theorem 4.5) and distinguish the words "invertible" and "isomorphic";
11. give a constructive proof that two spaces of the same finite dimension are isomorphic: prescribe $T(v_i) = w_i$ on bases, then use the dimension theorem for injectivity.

Exercises (printed pages). Lay: §2.1 p. 100–102 (Exercises 1–41: matrix arithmetic, the row–column rule, transposes and powers), §2.2 p. 109–111 (Exercises 1–42: inverses, elementary matrices, the inversion algorithm), §2.3 p. 115–116 (Exercises 1–45: applications of the Invertible Matrix Theorem); Friedberg: §2.3 p. 96 (Exercise 1 true/false, Exercise 8 the general form of Theorem 1.2, Exercise 18 a direct proof of associativity), §2.4 p. 106 (Exercise 3 deciding whether two spaces are isomorphic, Exercises 15–16 criteria for and construction of isomorphisms). The Practice Problems that precede each exercise block are worth doing first.

How to use the two texts together:

- **Computation and proof are split cleanly.** The three hand skills, matrix multiplication, inversion, elementary matrices, are in Lay (§2.1–§2.3); the three things that must be said exactly, composition, matrix representation, isomorphism, are in Friedberg (§2.3–§2.4). Lay's Figures 2 and 3 (p. 94) give the picture of matrix multiplication; Friedberg's Theorem 2.3 gives its proof.
- The spine of the lecture: *define composition* → *composition stays linear* → *define the matrix product* → $[UT] = [U][T]$ → *inverses and inverse matrices* ($[T^{-1}] = ([T])^{-1}$) → *isomorphic means "same dimension"*.
- The central result is Theorem 2.3: it translates "how maps compose" into "how matrices

multiply", and every later question about compositions becomes a question about matrix products. It rests on Theorem 2.2 (multiplying by a matrix is taking the coordinates of an image).